

Jimin Ryu

Email: jiminryu2@gmail.com | Website: jiminryu.com | linkedin.com/in/jryu0 | github.com/jiminleeryu

EDUCATION

Brown University, Providence, RI

Expected Graduation: **May 2026**

Computer Science – Applied Mathematics Sc.B.

GPA: 3.90/4.0

Relevant Coursework: Machine Learning, Software Engineering, Statistical Inference, Discrete Structures, Computer Systems, Linear Programming, Operating Systems, Data Structures and Algorithms, Linear Algebra

Skills: Java, Python, C/C++, Kotlin, GraphQL, JavaScript, HTML/CSS, MySQL, Assembly, React, Node.js, TensorFlow, PyTorch, Git, Docker, GCP, AWS, Visual Studio, PyCharm, Postman, Agile, Project Management, UI/UX

EXPERIENCE

Wingspans (Techstars '24 Startup), *Software Engineer Intern*

March 2024 – Present

Gatsby, Algolia, Cosmic.js, Figma, GraphQL, Jira

- Fully owned and launched automated email marketing tool for **50,000+ DAU**, personalized for user verification, weekly updates, and customer relationship management initiatives
- Developed modular shareable portfolio designer and career builder features, boosting user engagement by **85%**
- Engineered experience translator using natural language processing to tailor user testimonials to specific job paths

University of Idaho, *Research Assistant*

November 2023 – February 2024

Docker, REST, Pytorch, NumPy, TensorFlow, Linux

- Developed a Convolutional Neural Network model for binary classification using images collected by a custom-built unmanned aerial system, enhancing efficiency for agricultural practices
- Containerized the model with Docker, deployed it on a Linux virtual machine, and implemented a RESTful API using Flask to test and validate the model's performance on new image data
- Achieved **94%** accuracy and a testing loss of **0.00035** across 50,000 high-resolution images (1920x1080)

Full Stack At Brown, *Project Manager*

September 2024 – Present

React, Ruby, Jira

- Led 14 developers and designers to build platform and web server hosting 35 PhD students and 300+ publications
- Engaged with Brown's Physics Department chair and professors through biweekly discussions on project execution

Brown Department of Education, *Software Engineer*

September 2023 – January 2024

TypeScript, Java, Spark, MongoDB, Express

- Constructed a full-stack website to showcase the impact of 600+ parent-school relationships on student success
- Led backend development with query feature for handling, storage, and parsing of survey data

PROJECTS

SpotiFinder, *TypeScript, Java, Spark, Spotify API*

June 2024

- Developed a music discovery application leveraging Spotify API and regression models for artist recommendation
- Implemented k-nearest neighbors (k-NN) algorithm to curate a blended result of two separate artists trained over 3,000 artists' discography and song data across 13 attributes
- Built a tree-recursion algorithm to optimize music discovery by finding similar artists using popularity metrics

Interactive Maps Data Viewer, *TypeScript, Java, Spark, JUnit, Jest, Playwright*

May 2024

- Integrated Mapbox with GeoJSON overlay for interactive visualization of historical redlining data
- Developed a web API server with caching for efficient querying and viewing of data in CSV/JSON formats
- Implemented comprehensive testing in TypeScript and Java to ensure secure and reliable performance

Flight Controller Application, *Android Studio, Kotlin, Java, Jetpack Compose, C++*

August 2023

- Mobile development for intuitive Android interface for drone control using a hybrid Java and Kotlin stack
- Created packet communication protocol to control drone telemetry via BLE (Java encoding, C++ decoding)
- Implemented scalable Model-View-ViewModel architecture to enhance app reliability and maintainability

AIthello, *Java, JavaFX*

January 2023

- Created Othello game with three fully optimized computer player difficulties
- Implemented minimax with alpha-beta pruning, increasing prediction depth by evaluating future moves